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Essentials of Data Science  
Laboratory - 2304102L - 2026

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4.1. Practice Lab Assignment

4.1.1. Pandas - series creation and manipu...

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4.1.3. Student information

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4.1.1. Pandas - series creation and manipulation 03:51

Write a Python program that takes a list of numbers from the user, creates a Pandas series from it, and then calculates the mean of even and odd numbers separately using the `groupby` and `mean()` operations.

Hint: You can group the Series based on whether each number is even or odd.

**Input Format:**

- The user should enter a list of numbers separated by space when prompted.

**Output Format:**

- The program should display the mean of even and odd numbers separately.
- Each mean value should be displayed with a label indicating whether it corresponds to even or odd numbers.

Refer to the sample test cases for better understanding regarding input and output format.

Sample Test Cases

seriesMa...

```
1 # import pandas as pd
2
3 ## Take inputs from the user to create a list of
4 # numbers = list(map(int, input().split()))
5
6 ## Create a Pandas series from the list of numbers
7 # s = pd.Series(numbers)
8 ## Grouping by even and odd numbers and
9 # grouped = s.groupby(s % 2).mean()
10
11 ## Display the mean of even and odd numbers with
12 # labels
13 # grouped.index = ['Even' if is_even else 'Odd'
14 # for is_even in grouped.index]
15 # print("Mean of even and odd numbers:")
16 # print(grouped)
17
18 ## Display results
19 # for key, value in grouped.items():
20 #     if key == 0:
21 #         print(f"Mean of Even numbers: {value}")
```

Terminal Test cases

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11:36 10-05-2026

### Create the DataFrame:

- Convert the dictionary to a Pandas DataFrame.

Add a new row:

- Take inputs from the user for the new row data (name, age).
- Add the new row to the DataFrame.
- Display the DataFrame after adding the new row.

### Modify a row:

- Modify a specific row by changing the age. Take the row index and new age value from the user.
- Display the DataFrame after modifying the row.

**Delete a row:**

- Take the row index to be deleted from the user.
- Remove the specified row.
- Display the DataFrame after deleting the row.

Add a new column:

- Add a column **Gender** with values taken from the user.
- Display the DataFrame after adding the new column.

### Modify a column:

- Convert names to unnames

```
datafram...
1 # import pandas as pd
2
3 ## Provided dictionary of lists
4 # data = {
5 #     'Name': ['Alice', 'Bob', 'Charlie'],
6 #     'Age': [25, 30, 35],
7 # }
8
9 ## Convert the dictionary to a DataFrame
10 # df = pd.DataFrame(data)
11
12 ## Display the original DataFrame
13 # print("Original DataFrame:")
14 # print(df)
15
16 ## Adding a new row
17
18
19 ## Display the DataFrame after adding a new row
20 # print("After adding a row:\n",df)
21
22 ## Modifying a row
23
24
25 ## Display the DataFrame after modifying a row
```

## Pin

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- 4.2: Lab Assignment
- 5: Practical 5

## 08:01 AA ☾ ✎ 🔗 —

- Display the first five rows of the data frame.
- Calculate the average age of the students (limit the average age up to 2 decimal places).
- Filter out the students who have a grade above a certain threshold (consider the threshold grade is 'B').

#### Sample Test Cases

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4. Practical 4

4.1 Practice Lab Assignment

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4.2.1. Month with the Highest Total Sales

01:55

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

The CSV file contains the columns: Date, Product, Quantity, Price, and City.

Group the data by Month and calculate the total sales for each month.

Find the month with the highest total sales and display it.

Also, display the total sales for the best month.

Sample Data:

| Date       | Product   | Quantity | Price | City        |
|------------|-----------|----------|-------|-------------|
| 2025-01-01 | Product A | 5        | 20    | New York    |
| 2025-01-01 | Product B | 3        | 15    | Los Angeles |
| 2025-01-02 | Product A | 7        | 20    | New York    |
| 2025-01-02 | Product C | 4        | 30    | Chicago     |
| 2025-01-03 | Product B | 2        | 15    | Chicago     |
| 2025-01-03 | Product A | 8        | 20    | Los Angeles |
| 2025-01-04 | Product C | 6        | 30    | New York    |
| 2025-01-04 | Product B | 5        | 15    | Los Angeles |
| 2025-01-05 | Product A | 3        | 20    | Chicago     |
| 2025-01-05 | Product C | 10       | 30    | Los Angeles |

Note:

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

monthFor...

Submit

1

# import pandas as pd

2

3

## Prompt the user for the file name

4

# file\_name = input()

5

6

## Load the data

7

# df = pd.read\_csv(file\_name)

8

9

10

## Find the month with the highest total sales

11

# best\_month =

12

# highest\_sales =

13

14

# print(f"Best month: {best\_month}")

15

# print(f"Total sales: \${highest\_sales:.2f}")

16

import pandas as pd

17

18

## Prompt the user for the file name

19

file\_name = input()

20

21

## Load the data

22

df = pd.read\_csv(file\_name)

23

df['Date'] = pd.to\_datetime(df['Date'])

24

25

df['Month'] = df['Date'].dt.to\_period('M')

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4.2.2. Best Selling Product

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Find the product that sold the most in terms of quantity sold.
- Display the product that sold the most and the total quantity sold for that product.

Sample Data:

| Date       | Product   | Quantity | Price | City        |
|------------|-----------|----------|-------|-------------|
| 2025-01-01 | Product A | 5        | 20    | New York    |
| 2025-01-01 | Product B | 3        | 15    | Los Angeles |
| 2025-01-02 | Product A | 7        | 20    | New York    |
| 2025-01-02 | Product C | 4        | 30    | Chicago     |
| 2025-01-03 | Product B | 2        | 15    | Chicago     |
| 2025-01-03 | Product A | 8        | 20    | Los Angeles |
| 2025-01-04 | Product C | 6        | 30    | New York    |
| 2025-01-04 | Product B | 5        | 15    | Los Angeles |
| 2025-01-05 | Product A | 3        | 20    | Chicago     |
| 2025-01-05 | Product C | 10       | 30    | Los Angeles |

Note:  
The data cannot be displayed in the file. You can refer to the sample data provided for insights.

monthFor...

```
1 # import pandas as pd
2
3 ## Prompt the user for the file name
4 # file_name = input()
5
6 ## Load the data
7 # df = pd.read_csv(file_name)
8
9
10 ## Find the product with the highest total
    quantity sold
11 # best_product =
12 # highest_quantity =
13
14 ## Display the result
15 # print(f"Best selling product: {best_product}")
16 # print(f"Total quantity sold: {highest_quantity}")
17 import pandas as pd
18
19 # Prompt the user for the file name
20 file_name = input()
21
22 # Load the data
23 df = pd.read_csv(file_name)
24 product_sales = df.groupby("Product")
```

Terminal Test cases

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5. Practical 5

## 00:42 AA ☾ ☑ ☎ -

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Group the data by City and calculate the total quantity of products sold for each city.
- Find the city that sold the most products (based on the total quantity sold).

| Date       | Product   | Quantity | Price | City        |
|------------|-----------|----------|-------|-------------|
| 2025-01-01 | Product A | 5        | 20    | New York    |
| 2025-01-01 | Product B | 3        | 15    | Los Angeles |
| 2025-01-02 | Product A | 7        | 20    | New York    |
| 2025-01-02 | Product C | 4        | 30    | Chicago     |
| 2025-01-03 | Product B | 2        | 15    | Chicago     |
| 2025-01-03 | Product A | 8        | 20    | Los Angeles |
| 2025-01-04 | Product C | 6        | 30    | New York    |
| 2025-01-04 | Product B | 5        | 15    | Los Angeles |
| 2025-01-05 | Product A | 3        | 20    | Chicago     |
| 2025-01-05 | Product C | 10       | 30    | Los Angeles |

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

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4.2.4. Most Frequently Sold Product Pairs

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

The CSV file contains the following columns: Date, Product, Quantity, Price, and City.

For each date, find all pairs of products that were sold together (i.e., two products sold on the same date).

Output the product pair/s that was sold most frequently.

Sample Data:

| Date       | Product   | Quantity | Price | City        |
|------------|-----------|----------|-------|-------------|
| 2025-01-01 | Product A | 5        | 20    | New York    |
| 2025-01-01 | Product B | 3        | 15    | Los Angeles |
| 2025-01-02 | Product A | 7        | 20    | New York    |
| 2025-01-02 | Product C | 4        | 30    | Chicago     |
| 2025-01-03 | Product B | 2        | 15    | Chicago     |
| 2025-01-03 | Product A | 8        | 20    | Los Angeles |
| 2025-01-04 | Product C | 6        | 30    | New York    |
| 2025-01-04 | Product B | 5        | 15    | Los Angeles |
| 2025-01-05 | Product A | 3        | 20    | Chicago     |
| 2025-01-05 | Product C | 10       | 30    | Los Angeles |

Explanation:

Transactions:

2025-01-01: Product A, Product B

2025-01-02: Product A, Product C

2025-01-03: Product B, Product A

2025-01-04: Product C, Product B

frequentl...

1 # import pandas as pd

2 # from itertools import combinations

3 # from collections import Counter

4

5 ## Prompt user to input the file name

6 # file\_name = input()

7

8 ## Read data from the specified CSV file

9 # df = pd.read\_csv(file\_name)

10

11 ## write the code

12

13

14 ## Output the most frequent product

15 import pandas as pd

16 from itertools import combinations

17 from collections import Counter

18

19 # Prompt user to input the file name

20 file\_name = input()

21

22 # Read data from the specified CSV file

23 df = pd.read\_csv(file\_name)

24 data\_by\_date = df.groupby('Date')

25 ['Product'].apply(list)

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5. Practical 5

4.2.5. Titanic Dataset Analysis and Data Cleaning

00:41

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset. For each question, perform necessary data cleaning, transformations, and calculations as required.

1. Display the first 5 rows of the dataset.

2. Display the last 5 rows of the dataset.

3. Get the shape of the dataset (number of rows and columns).

4. Get a summary of the dataset (using `.info()`).

5. Get basic statistics (mean, standard deviation, etc.) of the dataset using `.describe()`.

6. Check for missing values and display the count of missing values for each column.

7. Fill missing values in the 'Age' column with the median age.

8. Fill missing values in the 'Embarked' column with the most frequent value (`mode()`).

9. Drop the 'Cabin' column due to many missing values.

10. Create a new column, 'FamilySize' by adding the 'SibSp' and 'Parch' columns.

The Titanic dataset contains columns as shown below,

|   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|
| P | S | P | N | S | P | T | F | C | E |
| a | u |   |   |   |   |   |   |   | m |
| s |   |   |   |   |   |   |   |   | b |

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1

# import pandas as pd

2

# import numpy as np

3

4

## Load the Titanic dataset

5

# data = pd.read\_csv('Titanic-Dataset.csv')

6

7

## 1. Display the first 5 rows of the dataset

8

9

10

## 2. Display the last 5 rows of the dataset

11

12

13

## 3. Get the shape of the dataset

14

15

16

## 4. Get a summary of the dataset (info)

17

18

19

## 5. Get basic statistics of the dataset

20

21

22

## 6. Check for missing values

23

24

25

## 7. Fill missing values in the 'Age' column

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5. Practical 5

4.2.6. Titanic Dataset Analysis and Data Cleaning - 2

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Create a new column 'IsAlone' which is 1 if the passenger is alone (FamilySize = 0), otherwise 0.

2. Convert the 'Sex' column to numeric values (male: 0, female: 1).

3. One-hot encode the 'Embarked' column, dropping the first category.

4. Get the mean age of passengers.

5. Get the median fare of passengers.

6. Get the number of passengers by class.

7. Get the number of passengers by gender.

8. Get the number of passengers by survival status.

9. Calculate the survival rate of passengers.

10. Calculate the survival rate by gender.

The Titanic dataset contains columns as shown below,

|   |   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|---|
| P | S | P | N | S | A | S | P | T | F | C | E |
| a | u | l | a | e | g | i | a | i | a | a | m |
| s | r | a | m | e | e | b | r | k | r | b | b |
| s | v | s | e |   |   |   | c | e |   |   |   |
| e | i |   |   |   |   |   | h | t |   |   |   |
| n | d |   |   |   |   |   |   |   |   |   |   |
| g |   |   |   |   |   |   |   |   |   |   |   |
| e |   |   |   |   |   |   |   |   |   |   |   |
| r |   |   |   |   |   |   |   |   |   |   |   |
| i |   |   |   |   |   |   |   |   |   |   |   |

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18

19

20

21

22

# import pandas as pd

# import numpy as np

## Load the Titanic dataset

# data = pd.read\_csv('Titanic-Dataset.csv')

# data['FamilySize'] = data['SibSp'] +

data['Parch']

## 1. Create a new column 'IsAlone' (1 if alone,

0 otherwise)

## 2. Convert 'Sex' to numeric (male: 0, female:

1)

## 3. One-hot encode the 'Embarked' column

## 4. Get the mean age of passengers

## 5. Get the median fare of passengers

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4.2.7. Titanic Dataset Analysis and Data Cleaning - 3

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Calculate the survival rate by class.

2. Calculate the survival rate by embarkation location (Embarked\_S).

3. Calculate the survival rate by family size (FamilySize).

4. Calculate the survival rate by being alone (IsAlone).

5. Get the average fare by passenger class (Pclass).

6. Get the average age by passenger class (Pclass).

7. Get the average age by survival status (Survived).

8. Get the average fare by survival status (Survived).

9. Get the number of survivors by class (Pclass).

10. Get the number of non-survivors by class (Pclass).

The Titanic dataset contains columns as shown below,

|   |   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|---|
| P | S | P | N | S | A | S | P | T | F | C | E |
| a | u | l | a | e | g | i | a | i | a | a | m |
| s | r | c | m | x | e | b | r | c | r | b | b |
| s | v | l | e |   |   |   | c | k | e | a | r |
| e | i | s |   |   |   |   | h | e | t | k | k |
| n | v |   |   |   |   |   |   |   |   | e | d |
| g |   |   |   |   |   |   |   |   |   |   |   |
| e |   |   |   |   |   |   |   |   |   |   |   |
| r |   |   |   |   |   |   |   |   |   |   |   |
| i |   |   |   |   |   |   |   |   |   |   |   |

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1 # import pandas as pd

2 # import numpy as np

3

4 ## Load the Titanic dataset

5 # data = pd.read\_csv('Titanic-Dataset.csv')

6 # data['FamilySize'] = data['SibSp'] +

7 # data['Parch']

8 # data['IsAlone'] = np.where(data['FamilySize'] >

9 0, 0, 1)

10

11 ## 1. Calculate the survival rate by class

12

13 ## 2. Calculate the survival rate by embarked

14 location

15

16 ## 3. Calculate the survival rate by family size

17

18

19 ## 4. Calculate the survival rate by being alone

20

21

Terminal Test cases

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5. Practical 5

## 00:43 🔍 🌙 📝 🔗 -

1. Get the number of survivors by gender (Sex).
2. Get the number of non-survivors by gender (Sex).
3. Get the number of survivors by embarkation location (Embarked\_S).
4. Get the number of non-survivors by embarkation location (Embarked\_S).
5. Calculate the percentage of children (Age < 18) who survived.
6. Calculate the percentage of adults (Age >= 18) who survived.
7. Get the median age of survivors.
8. Get the median age of non-survivors.
9. Get the median fare of survivors.
10. Get the median fare of non-survivors.

| Passenger Id | Survived | Class | Name | Sex | Age | SibSp | Parch | Ticket | Fare | Cabin | Embarked |
|--------------|----------|-------|------|-----|-----|-------|-------|--------|------|-------|----------|
|--------------|----------|-------|------|-----|-----|-------|-------|--------|------|-------|----------|

Submit

Terminal Test cases